Thermal fatigue life prediction and intermetallic compound behaviour of SAC305 BGA solder joints subject to accelerated thermal cycling test

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Abstract

Purpose – This study aims to investigate the thermal fracture mechanism of moisture-preconditioned SAC305 ball grid array (BGA) solder joints subjected to multiple reflow and thermal cycling.

Design/methodology/approach – The BGA package samples are subjected to JEDEC Level 1 accelerated moisture treatment (85 °C/85%RH/168 h) with five times reflow at 270 °C. This is followed by multiple thermal cycling from 0 °C to 100 °C for 40 min per cycle, per IPC-7351B standards. For fracture investigation, the cross-sections of the samples are examined and analysed using the dye-and-pry technique and backscattered scanning electron microscopy. The packages' microstructures are characterized using an energy-dispersive X-ray spectroscopy approach. Also, the package assembly is investigated using the Darveaux numerical simulation method.

Findings – The study found that critical strain density is exhibited at the component pad/solder interface of the solder joint located at the most distant point from the axes of symmetry of the package assembly. The fracture mechanism is a crack fracture formed at the solder's exterior edges and grows across the joint's transverse section. It was established that Au content in the formed intermetallic compound greatly impacts fracture growth in the solder joint interface, with a composition above 5 Wt.% Au regarded as an unsafe level for reliability. The elongation of the crack is aided by the brittle nature of the Au-Sn interface through which the crack propagates. It is inferred that refining the solder matrix elemental compound can strengthen and improve the reliability of solder joints.

Practical implications – Inspection lead time and additional manufacturing expenses spent on investigating reliability issues in BGA solder joints can be reduced using the study's findings on understanding the solder joint fracture mechanism.

Originality/value – Limited studies exist on the thermal fracture mechanism of moisture-preconditioned BGA solder joints exposed to both multiple reflow and thermal cycling. This study applied both numerical and experimental techniques to examine the reliability issue.

Keywords Solder joint, Moisture, Thermal cycling test, Reflow, Crack, Intermetallic compound

Paper type Research paper

1. Introduction

The recent advancements in the electronic industries, including integrating small-sized intricate electronic components for greater electronic functionality, have resulted in increased heat generation and inherent thermal fatigue, leading to reliability concerns (Apalowo *et al.*, 2023). A pertinent source of reliability concern in integrated packaging is the solder joints between the package's components (Chicot *et al.*, 2013). Solder joints assemble the package's components and provide signal transmission and structural supports among the components. Hence, they are subjected to thermal and mechanical fatigue over continuous device usage. A suitable method to assess the solder joints' reliability is subjecting them to an accelerated thermal cycling test (TCT) over an elongated duration (Apalowo *et al.*, 2018). The test subjects the joints to periodic thermal stress and fatigue typical of an electronic device under operating conditions. Understanding the thermal fatigue behaviour of solder joints when subjected to an

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accelerated TCT is imperative to improving the thermal reliability of the package assembly.

The thermal fatigue life of a structural package is a measure of its thermal reliability. Over the past decades, many researchers have proposed numerical models for thermal fatigue life prediction of solder joints. The numerical models can be categorized based on the damage criterion parameter used for the fatigue life prediction. The creep damage-based model, which applies the creep-fatigue deformation of the solder joint when exposed to elevated temperature and thermal cycling, has been applied for the fatigue life prediction of solder joints. Examples include the Syed and Knecht–Fox model (Zhang *et al.*, 2009) and the Manson strain partitioning model (Wong, 2019). Also, strain energy-based models have been proposed to predict the thermal fatigue life of solder joints using the hysteresis energy of the joints when exposed to thermal cycling. Examples of strain energy-based models include the Stowell (Stowell, 1966), Pan (Pan, 1994), Solomon and Torksdorf (Solomon and Tolksdorf, 1995) and Shi (Shi *et al.*, 1999) models. Another prediction model is the plastic strain-based model, which applies the plastic-fatigue strain of the joint as an indicative damage parameter for predicting the fatigue life of the joint when exposed to cyclic temperature changes. Examples of the models include Solomon, Engelmaier and Coffin–Manson models (Wang and Tang, 2019; Depiver *et al.*, 2020; Pan *et al.*, 2020). The damage-accumulation models have also been developed to investigate solder joints' thermal fatigue failure mechanism (Apalowo *et al.*, 2024; Che Ani *et al.*, 2018). The models have the potential to predict the initiation and propagation of damage within the joint. A notable example of the damage-accumulation model is the Darveaux reliability model, which estimates the reliability (life prediction) of solder joints based on the accumulated plastic strain energy density of the joint interfaces (Su *et al.*, 2019; Darveaux, 2000; Gao and Kwak, 2021; Gharaibeh, 2022). The Darveaux method has proven highly effective and is relevant to the recent advancements in finite element technology. A comprehensive state-of-the-art review of the recent advancement of solder joint fatigue life prediction is presented in Qiu *et al.* (2022) and Su *et al.* (2019).

Lead-free solders have good mechanical and wettability properties, making them sound alternatives for electronics applications for lead-based solders, which exhibit adverse environmental challenges. Despite their comparative advantages, lead-free solders face reliability concerns, including mechanical and thermal fatigue (Che Ani *et al.*, 2019; Zhang *et al.*, 2010). Studies by many researchers in the past decade have shown that the microstructural contents of lead-free solders determine their plastic and creep behaviour during the thermal cycling process (Lin and Chu, 2005; Wiese and Wolter, 2007). Studies have also established that their microstructural contents greatly impact the mechanical reliability of lead-free solder jois (Liang *et al.*, 2017; Pang *et al.*, 2004). The formation of Au-Sn-based intermetallic compound (IMC) at the interface of lead-free solder joint indicates the joint's good wetting and bonding properties. Still, the IMC layer's growth negatively impacts the solder joint's strength (Liang *et al.*, 2017). This is because the IMC strengthens the solder joint, but an increase in its thickness results in void formation, which acts as a crack initiator. Studies have demonstrated that voids reduce the deformation resistance of solder joints (Kanchanomai *et al.*, 2002; Lin and Chu, 2005).

Studies on the thermal fracture mechanism of moisture-preconditioned lead-free SAC305 ball grid array (BGA) solder joints are limited in the existing literature. Also, the impact of the preconditioned solder's microstructural contents on its fracture growth phenomenon is inexistent in the literature. Therefore, this study investigates the thermal fracture mechanism of moisture-preconditioned SAC305 solder joints subjected to multiple reflow and thermal cycling. The solders are preconditioned through an accelerated moisture treatment (85°C/85%RH/168 h) with five times reflow at 270°C, followed by multiple thermal cycling. For fracture investigation, the cross-sections of the samples are examined and analysed using the dye-and-pry technique and backscattered scanning electron microscopy (SEM). The packages' microstructures are characterized using an energy-dispersive X-ray spectroscopy (EDX) approach to analyse the impact of the microstructural contents on the fracture growth phenomenon. Also, the package assembly is numerically investigated using the Darveaux reliability method.

2. Experimental investigation

2.1 Test specimen

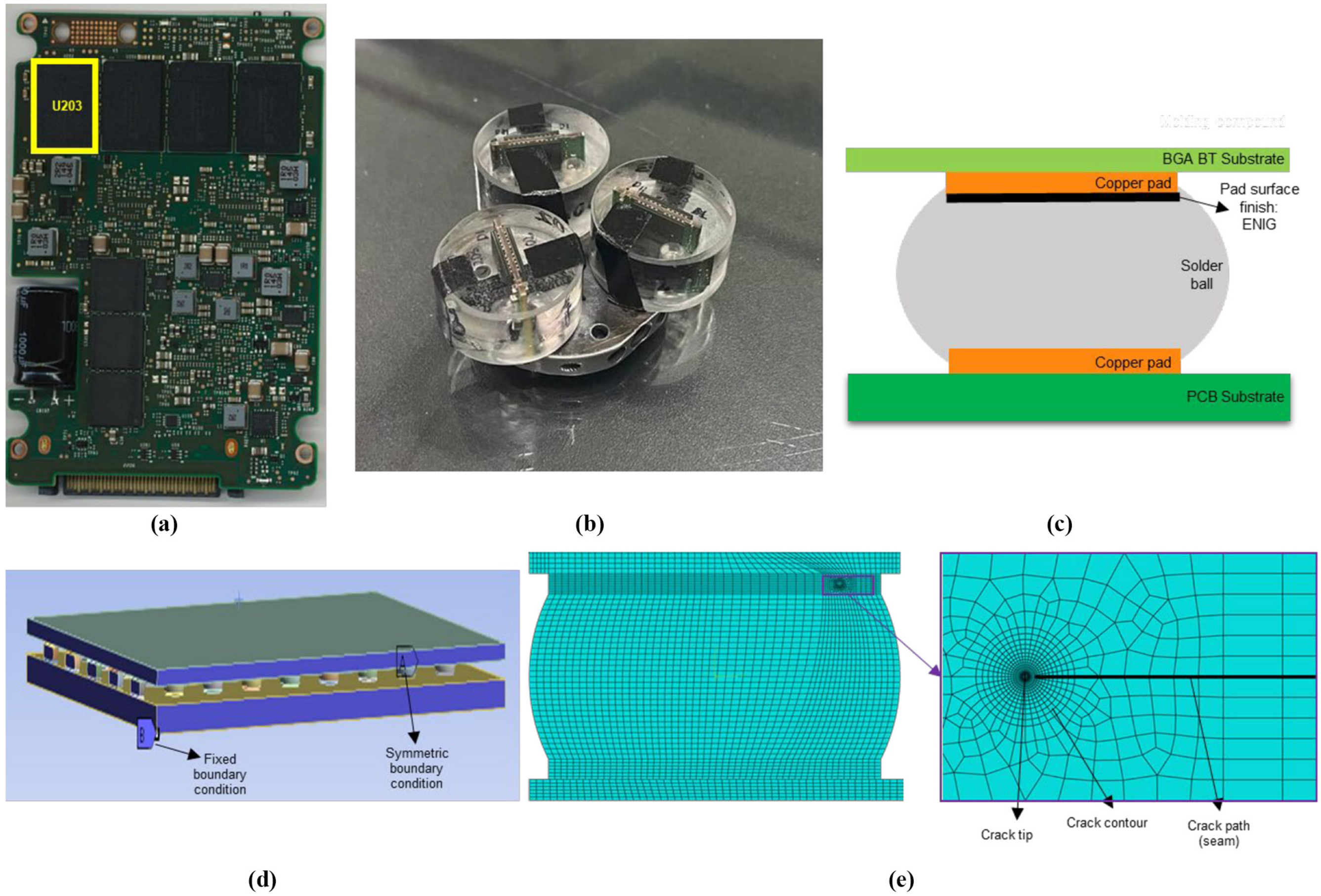
The test specimen used in the reliability investigation conducted in this study is Western Digital's U203 NAND BGA package assembly. The Western Digital's PCBA component, on which the U203 NAND is soldered, is shown in Figure 1(a). The NAND package (with an overall dimension of $18 \times 12 \times 1.55$ mm) consists of an array of 170 SAC305 (96.5Sn3.0Ag0.5Cu) balls sandwiched between a BT substrate and a copper pad on one side and a copper pad and a PCB substrate on the other side, as shown in Figure 1(c). The solder balls are of diameter 0.49 ± 0.05 mm with 1.0 mm pitch. The BT substrate, copper pads and PCB substrate are 0.175-, 0.025- and 0.3-mm thick, respectively. The copper pad on the component side has an ENIG (electroless nickel/immersion gold) surface finish.

The prepared samples [Figure 1(b)] are coated with carbon since carbon's X-ray peak does not conflict with the peak of any other microstructural element (as this work seeks to detect Au) during the EDX analysis. Coating of the samples during electron microscopy enables or improves the imaging of the samples. Creating a conductive metallic layer on the sample inhibits charging, reduces thermal damage and improves the secondary electron signal required for topographic examination in the SEM.

2.2 Reflow and thermal cycling tests

Harsh thermal fatigue tests, which are accelerated over a short duration, are conducted to investigate the reliability of the solder joint when subjected to thermal fatigue. Two tests are performed: thermal reflow and TCT. The test samples are subjected to moisture soak treatment to introduce the typical harsh moisture contamination, which initiates microvoids along the interfaces of the BGA package assembly. The samples are exposed to JEDEC's Level 1 moisture soak preconditioning at 85°C and 85%RH for 168 h, per IPC and JEDEC joint Standard of J-STD-020D.1 (IPC/JEDEC, 2008). After the moisture preconditioning, the samples are reflowed at 270°C peak temperature five times. The thermal reflow process consists of

Figure 1 (a) PCBA board, showing the U203 BGA under investigation; (b) experimental test samples for microscopy investigation; (c) components of the BGA package assembly; (d) FEA model for the reliability simulation; (e) crack tip mesh in the FEA reliability model



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the steps of preheating, thermal soaking, reflow and cooling. Figure 2(a) shows the temperature profile of the reflow process.

The second part of the thermal fatigue tests is the multiple thermal cycling of the reflowed BGA samples. The reflowed samples are subjected to a TCT from 0°C to 100°C for 40 min per cycle, per IPC-7351B Standard (IPC, 2010). The thermal cycling commences with a heating ramp from the ambient temperature (25°C) to the highest temperature (100°C), followed by a 10-min soaking before the commencement of the first cycle of the TCT test. As shown in Figure 2(b), the first cycle of the TCT commences with a cooling ramp from the highest temperature (100°C) to the lowest temperature (0°C). Each thermal cycle consists of a 10-min ramp time for heating and cooling and a 10-min soak time after every cooling or heating ramp, as shown in Figure 2(b).

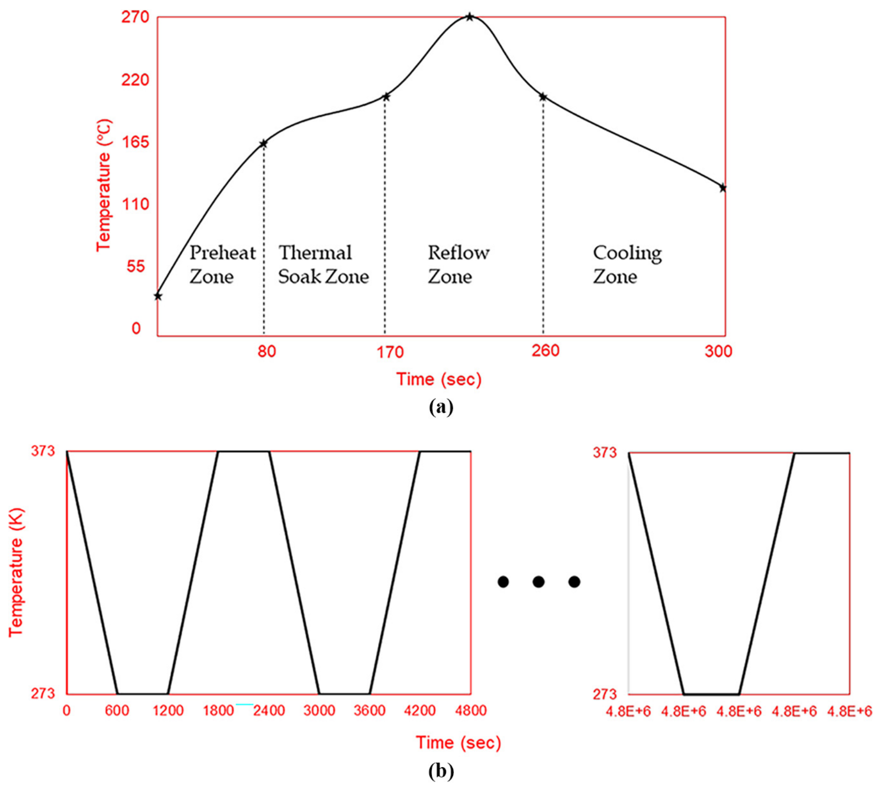
2.3 Microscopy inspections and analysis

Microscopy analyses, namely, backscattered SEM and EDX, are conducted to investigate and characterize the cross-sections of the solder joint assembly for thermal failure analysis. The post-reflowed BGA samples are inspected for crack growth using the ZEISS EVO SEM equipment with magnifications of 500× and 5000×. Also, the post-TCT samples are investigated using SEM to inspect crack growth and EDX analysis to characterize the microstructural behaviour of the solder joint at magnification 5000×. Observations of the thermally cycled samples are done at an initial interval of 50 cycles, then increased to an interval of 100 cycles after the crack has been fully formed. The thermal fatigue inspection data are collected, and reliability analysis is conducted on the solder joint packages.

3. Numerical simulation

Numerical simulations are performed to investigate thermal fatigue behaviour at the horizontal and vertical sections of the solder joint interface. The Darveuax reliability method analyses the horizontal X-section crack behavior at the package assembly's component pad/solder interface. Also, the finite element thermomechanical simulation is conducted to study the vertical X-section crack propagation along the package assembly's component pad/solder interface. The FE models are created and implemented using commercially available FEA software. A Dell PowerEdge Intel® Xeon 2.67 GHz Quad Core with 96 GB of RAM is used for the numerical simulations.

Figure 2 Temperature profiles of the (a) thermal reflow and (b) thermal cycling tests



Notes: (a) Thermal reflow; (b) TCT
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Figure 1(c) shows the FEA models' components. Table 1 presents the thermomechanical properties of the package assembly components, with thermal conductivity, density, Poisson's ratio, elastic modulus and coefficient of thermal expansion denoted as μ, ρ, υ, E and CTE, respectively. The crack growth correlation constants, which are typically computed experimentally, are obtained from relevant literatures (ANSYS Inc, 2018; Darveaux, 2000; Wang *et al.*, 2007).

3.1 Horizontal X-sectional thermal fatigue failure analysis

The viscoplastic finite element analysis (FEA) predicts the accumulated plastic strain energy density (APSED) and the solder joint's life in a flip-chip package assembly. The FEA modelling is based on the Darveaux reliability method (Che and Pang, 2004; Darveaux, 2000) for predicting the solder joint's life under thermal cycling. Darveaux reliability model is a damage-accumulation model which estimates the reliability (life prediction) of solder joints based on the accumulated plastic strain energy density imposed by the joint's gradual deformation when subjected to thermal cycling. The model computes the total fatigue life consists of the life associated with crack initiation and the one with crack growth. This allows the analysis of the crack growth mechanism of the solder joint. Other specific advantages of the model include the incorporation of microstructural evolution of the joint during

Table 1 Thermomechanical properties of BGA package components

Component	μ(W/m.K)	ρ(kg/m^3)	υ	E (GPa)	CTE (ppm/K)
PCB substrate	10	1,900	0.11	19	16
SAC305 solder	57.8	7,400	0.36	51 @0°C 44 @100°C	21 @ 0 °C 23.5 @ 100 °C
Cu pad	389	8,960	0.34	120	16.5
ENIG	99	–	–	–	13.3
BT substrate	10	2,000	0.21	23	14
Crack growth correlation constants					
K_1 (cyc.Pa^{-k2})	K_2	K_3 ($m.cyc^{-1}$.Pa^{-k4})		K_4	
1.53 × 10^{10}	−1.52	2.58 × 10^{-12}		0.98	

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thermal aging process, and the inclusion of creep effects, accounting for time-dependent deformation of solder joints.

In the Darveaux reliability prediction, three thermal cycles are adequate to determine the fatigue life of the solder joint. The fatigue life is associated with the APSED per cycle, the number of cycles to initiate a crack and the crack growth rate per cycle. The volume-averaged APSED ΔW_{ave} is calculated as follows:

$$\Delta W_{ave} = \frac{\sum \Delta W \cdot V}{\sum V} \tag{1}$$

where ΔW is the viscoplastic strain energy density accumulated per thermal cycle of each element, and V is the volume of each element. The number of thermal cycles before crack initiation N_0 is defined as follows:

$$N_0 = K_1 \cdot [\Delta W_{ave}]^{K_2} \tag{2}$$

where K_1 and K_2 are crack growth correlation constants of the solder joint. The crack growth rate per thermal cycle is computed as follows:

$$\frac{da}{dN} = K_3 \cdot [\Delta W_{ave}]^{K_4} \tag{3}$$

where K_3 and K_4 are crack growth correlation constants. The solder joint fatigue life is calculated as the number of thermal cycles before failure as follows:

$$N_f = N_0 + \frac{a}{\frac{da}{dN}} \tag{4}$$

where a is the diameter at the solder interface, which corresponds to the length the crack will propagate before completely separating the solder ball from the board or the package. The material constants $K_j(j = 1\ 4)$ used for the solder joint are shown in Table 1.

To take advantage of the package's symmetric construction and save computational time, only a quarter of its geometry [Figure 1(d)] is considered in the numerical simulations conducted in this section. As shown in Figure 1(d), a symmetry boundary condition is applied along the symmetrical layers of the package, and a fixed boundary condition is applied at the centre corner of symmetry.

3.2 Vertical X-sectional crack growth simulation

The decoupled thermomechanical simulation is conducted to investigate the crack growth mechanism along the vertical cross-section of the solder joint assembly. The decoupled simulation involves thermal analyses followed by mechanical analysis, in which the mechanical analysis uses the thermal analyses results as the fatigue load.

A local FE model [Figure 1(e)] of the critical solder joint assembly is considered for the crack growth mechanism along the vertical cross-section of the joint. Moisture contamination introduces voids in the joint's interface, leading to crack initiation during TCT. To implement this, an initial microcrack is created along the Cu pad/solder joint interface. The meshed package assembly is shown in Figure 1(e). The model is checked for mesh dependency, and it is ensured that the aspect ratio of all elements is close to 1. The crack tip is meshed with plane stress triangular (CPS3) elements, while the other parts of the model are meshed with plane stress quadrilateral (CPS4R) elements.

The thermal analysis defines the BGA package as a transient heat transfer model with five reflow steps followed by 2,000 TCT steps. The temperature profiles of the heat transfer steps are shown in Figure 2. The time-dependent results of the thermal analyses are exported to the mechanical model. In the mechanical analysis, the BGA package is similar to that of the thermal analyses, except for the inclusion of a microvoid crack initiator. The modeling process is also similar except for the crack modeling and thermal loading. The crack is defined in the assembly interaction module using the contour integral crack definition. A collapsed element side seam with a mid-side node parameter of 0.25 is defined for the crack path [Figure 1(e)]. A static general model is applied in the solution step module for the simulation times of 1500 s and 4,800,000 s, which simulate five reflow cycles and 2,000 thermal stress cycles, respectively.

4. Results and discussion

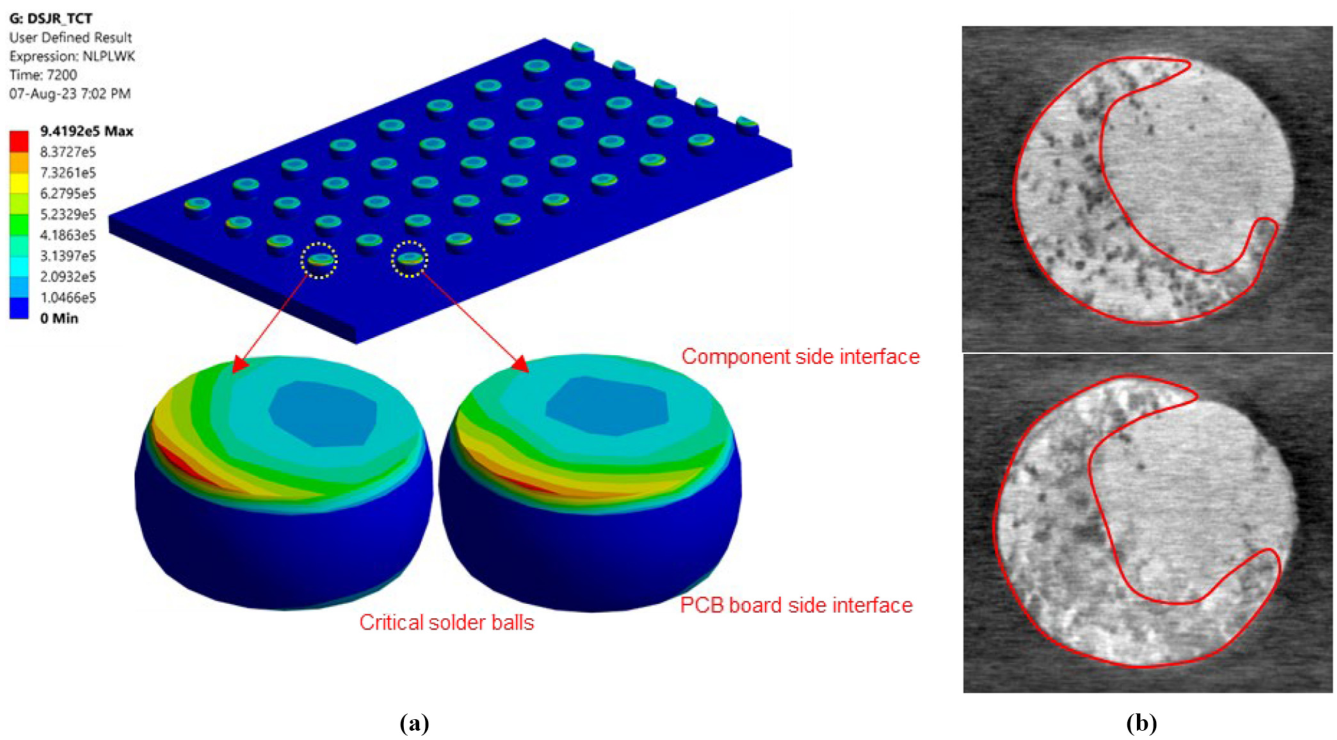
4.1 Dye-and-pry analysis of solder joint reliability and fatigue life prediction

The dye-and-pry failure analysis approach measures the crack propagation direction and failure mode in the moisture-preconditioned solder joint assembly. The dye-and-pry failure analysis predicts the thermal fatigue life of the package based on the creep strain imposed by the package's gradual deformation when subjected to thermal cycling. The predicted dye penetration (creep fracture behaviour) obtained through the Darveaux simulation is compared against the TCT experimental measurements, as shown in Figure 3. Figure 3 shows the horizontal cross-section of the solder joint subject to thermal cycling stress and the FEA prediction of the volume-averaged APSED. The FEA predictions of the crack initiation location and the crack growth direction agree with the experimental results, as shown in Figure 3. The maximum creep strain energy density occurs near the package's component side interface, as observed in the experimental result and FEA prediction. The critical value of APSED is observed in the component side interface of the solder ball joint (Figure 3) located at the most distant point from the axes of symmetry of the package assembly.

Figure 4 shows the volume-averaged APSED in the component and board interface of the critical ball per cycle for three thermal cycles. The volume-averaged APSED, which is the creep strain energy density of the interface, characterized the creep behaviour of the package assembly. The increment in the magnitude of the creep strain energy density during the ramp periods of thermal cycling is more significant than the cycling soak periods. As stated earlier, three thermal cycles are enough (Che and Pang, 2004) to obtain a stable volume-average APSED, which is applied to evaluate the fatigue life of the solder joint [equations (1) – (4)].

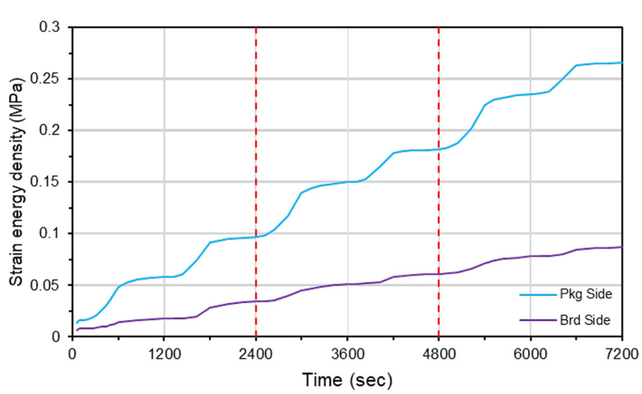
The crack initiation and ultimate failure cycles for the solder joint assembly at the package side interface are computed as 252 and 1,450 cycles, respectively. Meanwhile, crack initiation is observed after 2,520 cycles at the PCB board side interface.

Figure 3 Comparison of the horizontal X-sectional views of the solder joint interface obtained through thermal cycling experiment and FEA simulation



Notes: The mapped outline illustrates the dye-penetrated area of the joint interface; (a) FEA result; (b) dye penetration images
Source: Created by authors

Figure 4 Comparison of the accumulated creep strain energy density at the component and board sides interfaces



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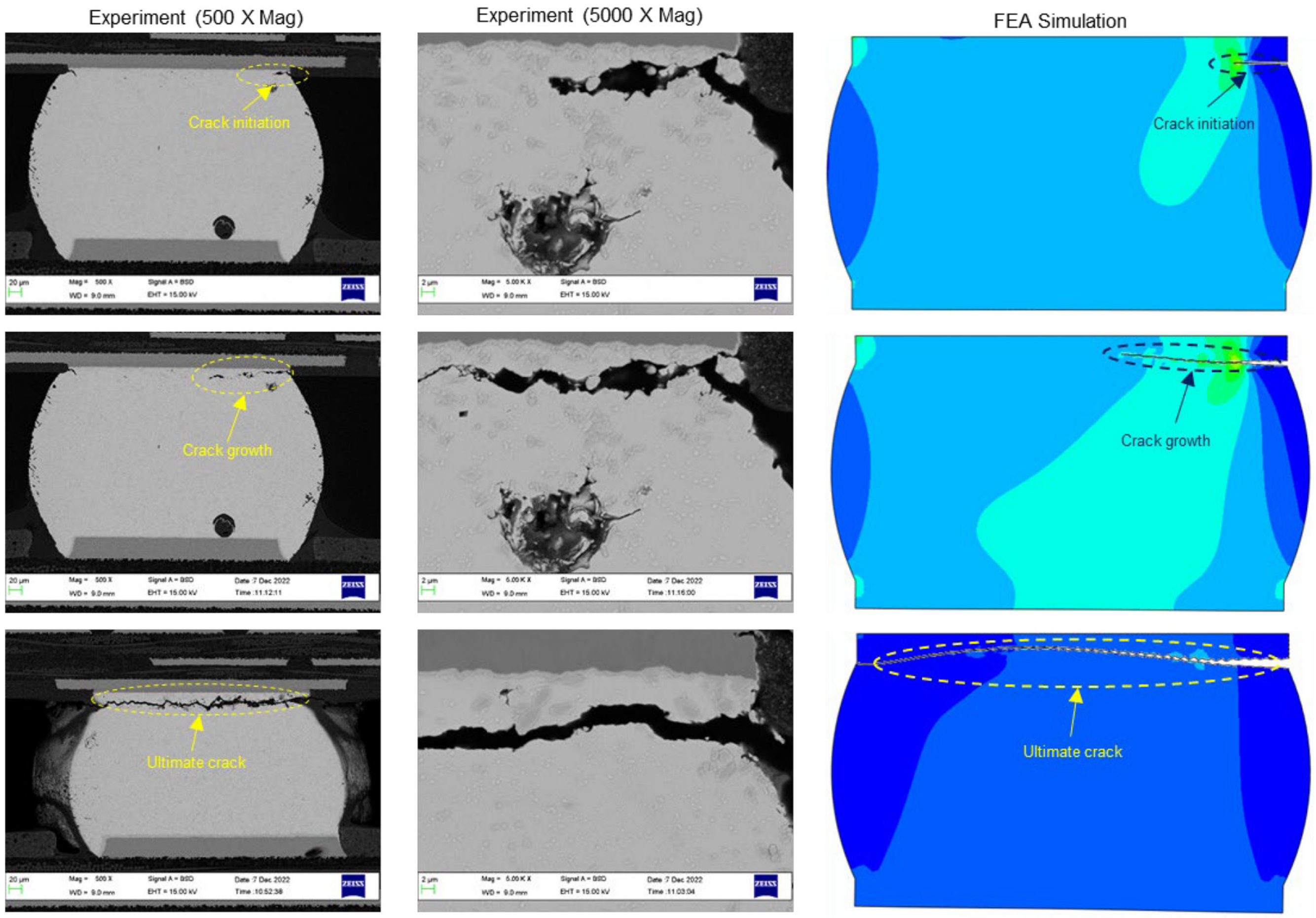
Also, as shown in Figure 4, the accumulated strain energy density observed at the component pad/solder interface (Pkg side) is higher than that of the PCB board/solder interface (Brd side). These observations signify that the package assembly is more prone to thermal reliability failure at the component pad/solder interface. The solder joint failure (crack) initiates when the joint interface is dyed up to 25% of the entire joint's area. Beyond this point, and with the continuous exposure of the package assembly to thermal stress, the dye extends until the joint ultimately fails.

4.2 Cross-sectional analysis of fracture growth in solder joint package assembly

Figure 5 shows the vertical X-sectional views of the moisture-preconditioned package assembly after thermal cycling, comparing the results of the thermal cycling experiment and the FEA simulation. SEM surface cross-sectional morphologies of the package sample are shown in 500× and 5000× magnifications.

Qualitatively, the FEA predictions of the crack initiation and propagation agree with the experimental observations of the crack growth, as shown in Figure 5. In both cases, cracks are observed at the component copper-pad/solder ball interface. The cracks show the tendency to extend along the cross-section of the solder ball. This indicates that the fatigue fracture of lead-free solder joints under the TCT is creep fracture, where the initiation always occurs in the external edges of solders, as shown in Figure 5. The cracks are more evident at the corners of the package side copper-pad/solder ball interface. The solder crack grows from the microvoid that is initiated by the thermal expansion of the joint interface during thermal cycling. The first appearance of crack occurs as a microvoid with lateral length in the range of 13 and 57 μm. As the TCT continues, the applied thermal stress tears open the initial crack (microvoid) along the direction of the stress and propagates the crack in the transverse direction (Ng *et al.*, 2023; Van Mier, 1991). The moisture preconditioning of the BGA package prior to the TCT can be attributed to the formation of the microvoid (crack initiator) (Lavoie *et al.*, 2007). During thermal cycling, vapor pressure is exerted

Figure 5 Comparison of the vertical X-sectional SEM images of the solder joint after the thermal cycling experiment against FEA predictions



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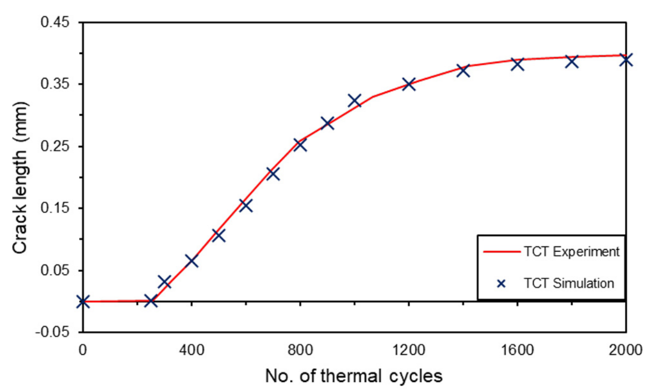
on the moisture-preconditioned joint, leading to the formation of microvoid. The microvoid, formed by exerted vapor pressure, acts as a crack initiator and propagates inwards during the thermal expansion of the package.

Another possible agent of the crack growth can be attributed to the significant difference in the thermal expansion coefficients (CTE) of the component layers (i.e. solder ball, copper pad, and ENIG) at the substrate side of the package assembly. While the formed crack propagates along the solder filler layer, it cuts through the component layers during crack growth. Specifically, the CTEs of the copper pad and ENIG are lower than those of the solder ball, leading to an irregular deformation rate during the thermal expansion of the assembly package. The irregular deformation thus results in a lower resistance to fracture growth at the assembly's component pad/solder ball interface.

The fracture growth during the thermal cycling process is shown in Figure 6, comparing the results obtained through the experiment and the FEA simulation. The evaluation compares the crack length growth rates of the two methods. Quantitatively, the FEA predictions of the crack growth agree with the experimental observations, with a maximum disparity of 1.57%. The number of cycles to the first crack initiation is recorded as 265 cycles, with the crack length observed in the range of 13 and 57 μm. After initiation, the crack continues to grow at an initial rate of 0.1 mm per 250 cycles. The growth rate is arrested when the crack length reaches about 0.35 mm. The ultimate failure, with a crack which runs through the entire

Figure 6 Fracture (crack length) growth during thermal cycling process: TCT experiment vs TCT simulation



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cross-sectional length of the solder joint interface, is observed at thermal cycling of around 1,500 cycles.

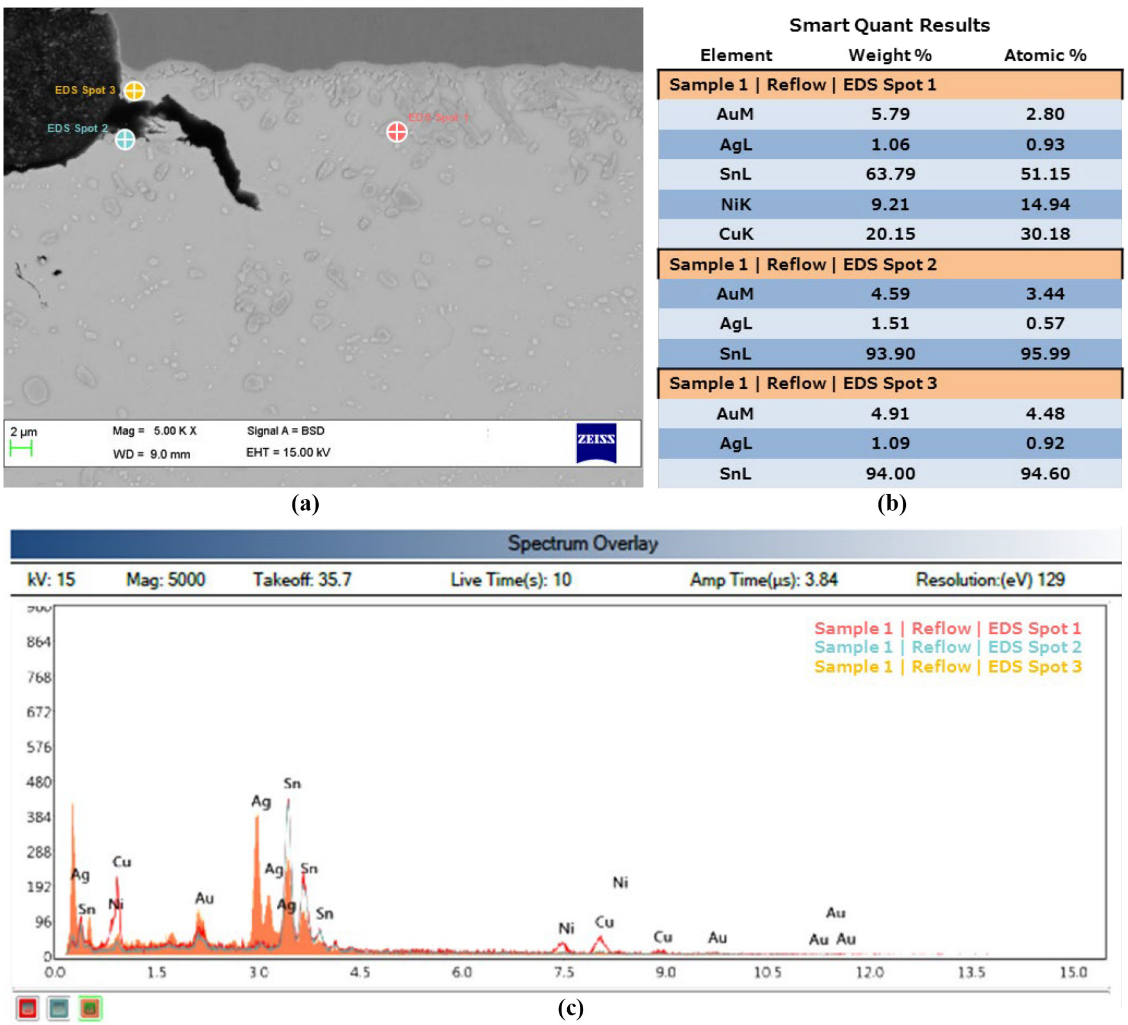
4.3 Microstructural characterization of the solder joint post-reflow and post-thermal cycling test

4.3.1 Microstructural analysis of the solder joint after thermal reflow

Figure 7 shows the FESEM micrographs and the EDS analysis of the intermetallic compounds formed at the solder joint after multiple reflow at 270°C for five cycles. Different spots around the crack region are investigated, as shown in Figure 7(a). As shown in the SEM images, the fracture mechanism is a crack that initiates from the solder matrix's corner. It tends to propagate along the solder matrix, at about 3–5 μm below the solder joint interface.

As shown in the backscattered SEM images [Figure 7(a)], the Au-Sn IMC is distributed across the solder matrix of the reflowed solder joint. Based on the EDS analyses of different spots of the solder joint assembly, it is found that different amounts of Ag, Ni and Cu atoms substitute the Au-Sn IMC sublattice particles across the crack region at the solder joint interface and within the solder matrix. From the analyses, three forms of sublattice particle substitutions are identified. Across the crack region, Ag substitution with no Cu and Ni detected are observed. On the contrary, within the solder matrix, Ag, Cu and Ni substitution of the Au-Sn sublattice particles is observed. These outcomes suggest that no specific composition of Au-Sn IMC is distributed across the solder matrix of the reflowed solder joint assembly. As shown in the EDS analysis [Figure 7(b)], the solder joint exhibits (Au, Ag) Sn IMC around the crack region, with 4.59–4.91 Wt.% Au, 1.09–1.51 Wt.% Ag and 93.9–94 Wt.% Sn. Meanwhile, above the crack region near the component side of the solder joint, (Au, Ag, Ni, Cu) Sn IMC is exhibited, with 5.79 Wt.% Au, 1.06 Wt.% Ag, 20.15 Wt.% Cu, 9.21 Wt.% Ni and 63.79 Wt.% Sn. From these observations, it can be deduced that the solder joints' thermal and mechanical strength weakens along the brittle Au-Sn layer, leading to crack growth. The small amounts of Au (about 2.80–4.48 At.%) and Ag (about 0.57–0.93 At.%) in the IMC compounds in the solder joint indicate that only small amounts of the elements are

Figure 7 Intermetallic compounds formed at the BGA solder joint after multiple reflow



Smart Quant Results

Element	Weight %	Atomic %
Sample 1 | Reflow | EDS Spot 1		
AuM	5.79	2.80
AgL	1.06	0.93
SnL	63.79	51.15
NiK	9.21	14.94
CuK	20.15	30.18
Sample 1 | Reflow | EDS Spot 2		
AuM	4.59	3.44
AgL	1.51	0.57
SnL	93.90	95.99
Sample 1 | Reflow | EDS Spot 3		
AuM	4.91	4.48
AgL	1.09	0.92
SnL	94.00	94.60

Notes: (a) FESEM micrograph; (b, c) EDS analysis
Source: Created by authors

involved in the IMC formation. In contrast, the remaining significant amounts remained as Au-Sn compound in the post-reflowed solder joint. This finding denotes a slight decrease of the Au-Sn fraction in the solder matrix after multiple reflow. This explains why the crack growth mechanism propagates through the Au-Sn IMC of the reflowed solder joint (Pan *et al.*, 2011), as shown in Figure 7(a).

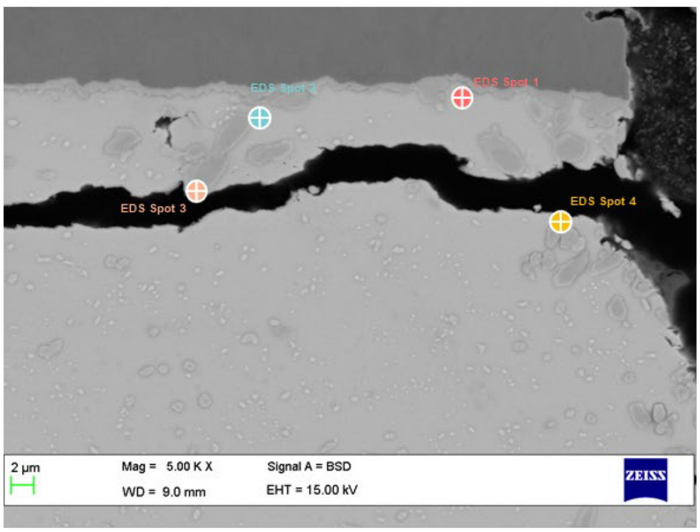
Therefore, based on these findings and discussions, it can be inferred that multiple reflows of the moisture-preconditioned BGA package induce the formation of Au-Sn IMC at the solder joint interface, which acts as crack initiator and propagates from the corner of the solder matrix during multiple reflows.

4.3.2 Microstructural analysis of the solder joint after thermal cycling

Figure 8 presents the FESEM micrographs and EDS analyses of the samples of the moisture-preconditioned BGA packages subject to 1,200 thermal cycles from 0°C to 100°C at 40 mins per cycle. Different spots around the crack region and the solder matrix are investigated, as shown in Figure 8(a). As shown in the SEM images, the failure mechanism is a crack propagating through the entire cross-sectional length of the solder joint interface.

Based on the EDS analyses of different spots of the thermally cycled solder joint assembly, it is found that different amounts of Ag and Cu atoms substitute the Au-Sn IMC sublattice particles across the crack region at the solder joint interface and within the solder matrix. Across the crack region, only Ag substitution with no Cu detected is observed. Meanwhile, in the solder matrix, Ag and Cu substitution of the Au-Sn sublattice particles is observed. Similar to the reflow case, these outcomes suggest that no specific composition of Au-Sn IMC is distributed across the solder matrix of the preconditioned

Figure 8 Intermetallic compounds formed at the BGA solder joint after thermal cycling

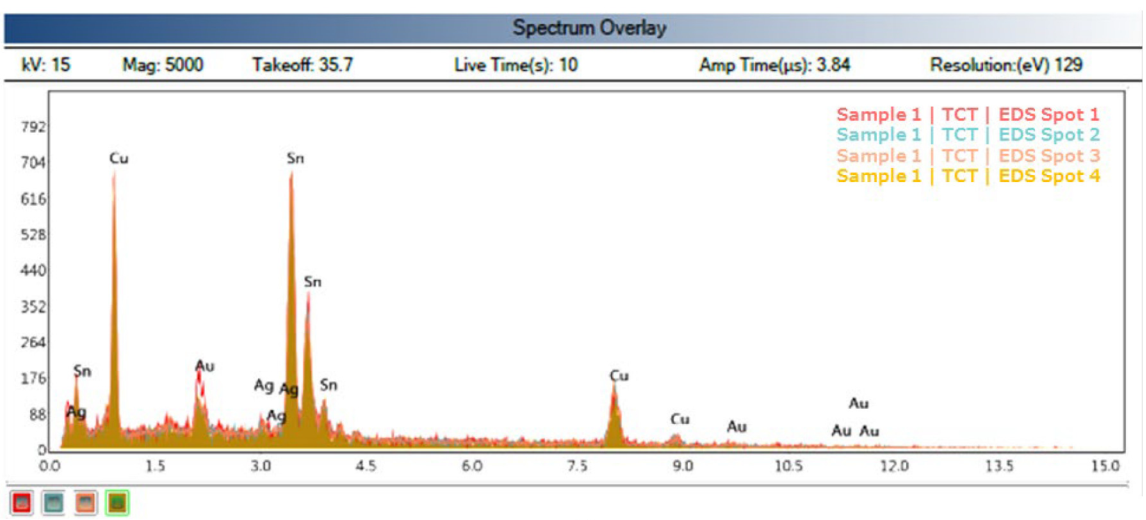


(a)

Smart Quant Results

Element	Weight %	Atomic %
Sample 1 | TCT | EDS Spot 1		
AuM	4.62	2.81
AgL	1.07	1.98
SnL	80.39	80.99
CuK	13.92	14.22
Sample 1 | TCT | EDS Spot 2		
AuM	5.84	3.60
AgL	0.55	0.62
SnL	78.90	83.23
CuK	14.71	12.55
Sample 1 | TCT | EDS Spot 3		
AuM	11.18	9.69
AgL	2.45	2.50
SnL	86.37	87.81
Sample 1 | TCT | EDS Spot 4		
AuM	12.04	10.92
AgL	2.60	2.54
SnL	85.36	86.54

(b)



(c)

Notes: (a) FESEM micrograph; (b, c) EDS analysis
Source: Created by authors

solder joint assembly subject to TCT. As shown in the EDS analysis [Figure 8(b)], the solder joint exhibits (Au, Ag)Sn IMC around the crack region, with 11.18–12.04 Wt.% Au, 2.45–2.60 Wt.% Ag and 85.36–86.37 Wt.% Sn. Meanwhile, above the crack region near the component side of the solder joint, (Au, Ag, Ni, Cu) Sn IMC is exhibited, with 4.62–5.84 Wt.% Au, 0.55–1.07 Wt.% Ag, 13.92–14.71 Wt.% Cu and 78.90–80.39 Wt.% Sn. Compared to the post-reflow results, it is observed that a substantial increase in the amount of Au occurs in the intermetallic compounds along the crack at the solder joint interface during the TCT. Au content increases from 4.59–4.91 Wt.% to 11.18–12.04 Wt.% and from 3.44–4.48 At.% to 9.69–10.92 At.% in the IMC formed along the solder joint crack path. The significant increase of Au content (and hence the reduction of other elements) during thermal cycling validated the migration of the elements towards the solder joint interface, leading to the increase in the formation of the unwanted Au-based IMC, which creates an embrittlement issue in the Sn-rich phase of the solder joint and resulting in the propagation of the already-initiated crack (Liang *et al.*, 2017). This phenomenon culminated in the increment of Au-Sn IMC at the solder joint interface of the thermally cycled package, leading to a crack growth mechanism that propagates through the solder joint's Au-Sn IMC (Pan *et al.*, 2011).

Therefore, based on these findings and discussions, it can be inferred that multiple reflow of the moisture-preconditioned BGA package induces the formation of Au-Sn IMC at the solder joint interface, which acts as a crack initiator and propagates inwards during thermal cycling.

5. Conclusion

This study investigates induced thermal fracture in moisture-preconditioned ball grid array (BGA) solder joints undergoing multiple reflow and thermal cycling. BGA samples undergo JEDEC Level 1 accelerated moisture treatment (85°C/85%RH/168 h) with five 270°C reflows, followed by thermal cycling (0°C to 100°C, 40 min per cycle, per IPC-7351B). Fracture analysis uses dye-and-pry technique and backscattered SEM, with microstructures characterized using EDX. The Darveaux reliability simulation is also used to numerically examine the package assembly.

Results reveal critical strain density at the component pad/solder interface furthest from the assembly's axes of symmetry. Failure manifests as crack fractures at solder exterior edges, growing transversely. Thermal failure is attributed to voids from moisture preconditioning or thermal mismatch. (Au, Ag)Sn intermetallic compound (IMC) forms along the crack path during reflow and thermal cycling, while (Au, Ag, Ni, Cu)Sn IMC forms in the solder matrix. Thermal fracture initiates at 57 μm size with Au content below 5 Wt.%, elongating when Au content reaches about 12 Wt.%, aided by the brittle Au-Sn interface. Au content exceeding 5 Wt.% is deemed unsafe for reliability. Cu inhibits Au-based IMC formation away from the crack path. Conclusively, refining solder matrix composition strengthens joints. More insights into fatigue life prediction can be established by investigating the impacts of thermal load and cycling temperature profile on the fatigue life of solder joint subjected to different accelerated thermal cycling loads. This is a potential topic for further research.

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